

2D ISAR Modified Orthogonal Matching Pursuit for Doppler-Ambiguous and Off-grid Target Scatterers

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ABSTRACT

Inverse Synthetic Aperture Radar (ISAR) enables imaging in any weather, day or night, but relies on the motion of a non-cooperative target to form an image. The crossrange resolution is dependent on angular diversity and is susceptible to aliasing if the pulse repetition frequency (PRF) is insufficient to process the incoming Doppler bandwidth of the received echoes. Doppler aliasing injects ghost scatterers into the unambiguous Doppler region which can be difficult to differentiate from authentic scatterers. Our proposed solution, Doppler Ambiguity Resolving Orthogonal Matching Pursuit (DAR-OMP), is a variant of OMP capable of disambiguating target scatterers through the target’s non-uniform motion. DAR-OMP identifies aliased targets in the unambiguous region and determines the originating ambiguity by maximizing the correlation of the residual with a localized neighborhood of candidate, ambiguous atoms. Additionally, DAR-OMP handles off-grid scatterers by applying Newton’s method with backtracking to refine the off-grid positions of the most probable candidate atoms. DAR-OMP is computationally efficient and scales easily by processing the latent image in a singular ambiguity. Preliminary results demonstrate as much as an 87.4% reduction in root mean squared error (RMSE) in comparison to Newtonized OMP with an expanded latent image containing the additional Doppler ambiguities. DAR-OMP is a light-weight, fast OMP variant capable of disambiguating target scatterers from different Doppler ambiguities by leveraging the non-uniform motion of a non-cooperative target.

Keywords: ISAR, doppler ambiguity, non-uniform motion, off-grid

1. INTRODUCTION

Inverse synthetic aperture radar (ISAR) imaging enables image creation in all weather at any time of the day or night. The range resolution of an ISAR image is dependent on the bandwidth of the radar, and the crossrange resolution is dependent on the angular diversity of the target. As the target rotates and the integration angle increases or the radar has more looks at a target, the crossrange resolution improves and scatterers are more easily differentiated. However, for non-cooperative targets, the target’s motion is typically complex and has high-order motion parameters, rotational acceleration, jerk, etc [1]. This complex motion can result in Doppler aliasing. Doppler aliasing or Doppler ambiguity arises when the PRF is less than twice the Doppler frequency of a scatterer. This is a common occurrence in applications like space surveillance and imaging of group targets where the far distances necessitates a low PRF or the extension of the target exceeds the Doppler bandwidth [2, 3]. The result is ambiguous Doppler frequencies fold into the unambiguous region where Doppler frequencies originating from ambiguous and unambiguous regions can no longer be differentiated.

In SAR imaging, there are a variety of different well-known and well-researched methods for handling Doppler ambiguity of a Doppler centroid. These methods are valuable but Doppler ambiguity arises differently in ISAR compared to SAR, and alternative ISAR-specific Doppler ambiguity resolver methods are worth exploring.

However, literature directly investigating resolving or removing ISAR Doppler ambiguity for extended or group targets is more scarce. The available literature is primarily focused on leveraging migration through range cells (MTRC) in order to differentiate ambiguous and unambiguous scatterers. The authors of [2] applying

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OMP to form an image by removing the Doppler ambiguity and correcting range cell. Their method leverages the differences in phase progression resulting from range-azimuth coupling and applies OMP to each range cell selecting Fourier-based atoms as a function of Doppler frequency, chirp rate and Doppler ambiguity. Using the selected atoms and subsequently estimated signal parameters, their method removes the Doppler ambiguity and corrects for range cell migration. This method effectively removes Doppler ambiguity; however, their method assumes uniform motion, the scatterers are on-grid, and their method relies on range-azimuth coupling in order to disambiguate atoms. Additionally, because the Doppler ambiguity is included as a dimension in their sensing matrix, the dimensionality and complexity far exceeds our proposed method. The authors of [3] use maximum weighted contrast to compare the range-walk produced by the estimated motion parameters and the range-walk for each ambiguity. The correct ambiguity corrects the range-walk and the scatterer’s peak sharpens. [3] applies a Gaussian weighting to mitigate the sidelobes of the echo where the standard deviation is proportional to the standard deviation of the data in each range cell [3]. Their method is an effective approach in addressing Doppler ambiguity, but assumes each range cell contains a singular scatterer and suffers from on-grid approximation in determining position.

Another assumption common in both [2,3] is the scatterers fall on-grid, dependent on the range and azimuth sampling. However, real targets do not fall perfectly inline with pre-defined grids, and this discrepancy can cause distortion errors resulting from basis mismatch [4]. The obvious fix is to increase the number of grid points, thereby distributing the energy of the measured reflectivity in a more exact position. This method has drawbacks, and is typically limited by both the additional computation required by increasing the dimensionality of the problem and by the increase in the coherence of atoms if using CS methods. Modern off-grid imaging methods like sparse Bayesian learning (SBL) [5–8] and atomic norm minimization (ANM) [9–12] are inherently continuous, enabling state-of-the-art reconstruction. However, SBL and ANM are computationally expensive and are not feasible for on-board image processing. In the pursuit of a real-time, off-grid ISAR image former, several authors have modified OMP. Most notably, Newtonized OMP (NOMP) cyclically refines each estimated target scattering position individually using Newton’s method [13]. Parameter-Refined OMP (PROMP) jointly refines the target scattering positions and reflectivities by calculating a Jacobian of the combined state [14].

1.1 Contributions

Traditionally higher-order motion parameters are seen as a nuisance as they can corrupt and contaminate the image formation process with unwanted phase errors. Higher-order motion parameters also introduce non-uniform phase progression and migration through range cells (MTRC). The goal of this paper is to leverage non-uniform phase progression from non-uniform motion to differentiate unambiguous and ambiguous target scatterers. Additionally, our formulation of the problem naturally enables off-grid scatterer compensation where we apply Newton’s method to a local neighborhood surrounding a candidate atom. Our key contributions are:

1. Our proposed method, AROMP is capable of differentiating Doppler ambiguous scatterers by leveraging the non-uniform phase progression resulting from the non-uniform rotation of the target
2. We perform off-grid reconstruction using Newton’s method within a neighborhood of ambiguous and un-ambiguous candidate atoms
3. We define a performance metric by which we can compare the outputs of ISAR images with strong individual scatterers.

2. BACKGROUND

ISAR imaging requires translational motion compensation and rotational motion compensation. Translational motion compensation is typically regarded as trivial and does not add anything outside of undesirable blurring. Assuming the translational motion compensation has been appropriately applied, Figure 1 depicts the geometry of the target during ISAR imaging where the target is rotating between pulses.

2.1 Target Geometry

The crossrange resolution of ISAR images is dictated by the rotation of the target. Assuming translational motion compensation has been applied, this reduces the ISAR model to a tabletop, where the target rotates at a fixed position away from the radar. Additionally, for the purposes of specifically evaluating the effects of non-uniform sampling and slow-time undersampling, the rotation motion parameters are assumed to be known. Figure 1 depicts the ISAR imaging geometry.

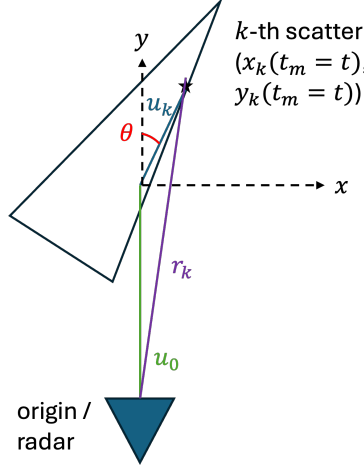


Figure 1. ISAR target geometry as a function of slow-time

The relative position of a scattering point as a function of slow-time (t_m) can be defined as [15]:

$$\begin{bmatrix} x_k(t_m) \\ y_k(t_m) \end{bmatrix} = \begin{bmatrix} \cos(\theta(t_m)) & -\sin(\theta(t_m)) \\ \sin(\theta(t_m)) & \cos(\theta(t_m)) \end{bmatrix} \begin{bmatrix} x_k \\ y_k \end{bmatrix} \quad (1)$$

where $\theta(t_m)$ or θ_m is the angle of the target scatterer relative to the y -axis, and $[x_k, y_k]^T$ is the initial position of the k -th scatterer relative to the center of the rotation of the target. The range from the origin located at the radar to a scattering point is [15]:

$$\begin{aligned} r_k(t_m) &= \sqrt{x_k^2(t_m) + (y_k(t_m) + u_0)^2} \\ &\approx u_0 + y_k(t_m) \quad \text{as } (y_k(t_m) + u_0)^2 \gg x_k^2(t_m) \\ &= u_0 + x_k \sin(\theta(t_m)) + y_k \cos(\theta(t_m)) \\ &= u_0 + u_k \\ &\text{where } u_k = x_k \sin(\theta(t_m)) + y_k \cos(\theta(t_m)) \end{aligned} \quad (2)$$

The angle θ_m can be approximated using a Taylor series:

$$\begin{aligned} \sin(\theta) &= \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \\ \cos(\theta) &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots \end{aligned} \quad (3)$$

For most ISAR applications, the target's rotation is relatively small across the coherent integration time, enabling the usage of the small angle approximation: $\sin(\theta(t_m)) \approx \theta(t_m)$ and $\cos(\theta(t_m)) \approx 1$. The range from the radar to a target scatterer can be approximated as:

$$\begin{aligned} r_k(t_m) &\approx u_0 + y_k + x_k \theta_m \\ \theta(t_m) &= \theta(0) + \omega t_m + \frac{\dot{\omega}}{2} t_m^2 + \dots \end{aligned} \quad (4)$$

2.2 Signal Model and Sensing Matrix

The signal can be represented as a range-compressed echo:

$$s(\hat{t}, t_m) = C \cdot \text{sinc}(B(\hat{t} - \tau(t_m))) \cdot \exp(-j2\pi f_c \tau(t_m)) \quad (5)$$

where C is the amplitude, B is the bandwidth, $\hat{t}$ is the fast-time, t_m is the slow-time, τ is the propagation delay and f_c is the center frequency. The propagation delay for a monostatic radar configuration is the amount of time required to propagate to the target and return:

$$\tau_m = \frac{2r(t_m)}{c} \quad (6)$$

where $r(t_m)$ is the range or distance from the radar to a target scattering point as defined by equation 2. The return echo can be represented as a summation of the returns from each scatterer:

$$s(\hat{t}, t_m) = C \sum_k \sigma_k \cdot \delta(\hat{t} - \tau_k(t_m)) \cdot \exp(-j2\pi f_c \tau_k(t_m)) \quad (7)$$

where k is the index of each scatterer located at $\delta(\hat{t} - \tau_k(t_m))$, K is the total number of scatterers or sparsity, and σ_k is the complex reflectivity of the k -th scatterer. Typical formulations will assume separability in the range and crossrange to simplify the sensing matrix's formulation; however, this approach neglects the cross-terms inherent to range-cell migration. To preserve this effect while still defining a linear system suitable for reconstruction using compressed sensing, apply the Fourier transform in the fast-time:

$$S(\hat{f}, t_m) = C \sum_k \sigma_k \cdot \exp(-j2\pi(f_c + \hat{f})\tau_k(t_m)) \quad (8)$$

where $\hat{f}$ is the temporal-frequency or the range-frequency. Equation 8 enables a formulation of a sensing matrix where each entry can be defined as:

$$A[m + \ell M, k] = C \sigma_k \cdot \exp(-j2\pi(f_c + \hat{f}_\ell)\tau_k(t_m)) \quad (9)$$

The resulting linear system is defined as: $y = Ax + n$, where $y \in \mathbb{C}^{ML}$ is the measurement, $A \in \mathbb{C}^{ML \times K}$ is the sensing matrix, and $x \in \mathbb{C}^K$ is the vectorized latent image. The dimensionality of the system is determined by the number of range-frequency samples L , the number of scatterers K and the number of received pulses M which is less than the number of total pulses N such that $M \ll N$. Figure 2 depicts the dimensionality of the proposed sensing matrix.

2.3 OMP

This system is underdetermined and has infinitely many solutions, each existing in the $(N - M)$ -dimensional null space of the sensing matrix. However, to treat this as a compressed sensing problem, if the latent image can be assumed K -sparse or containing K non-zero values, the problem becomes K -dimensional where $M \geq K$. Additionally, if the sensing matrix is roughly orthogonal and incoherent, associated with the Restricted Isometric Property (RIP), the solution can be unique [16]. The compressed sensing problem can then be formulated as:

$$\begin{aligned} & \min \|x\|_0 \quad \text{subject to} \quad \|y - Ax\|_2^2 \leq \text{constant} \\ \text{or} \quad & \min \|y - Ax\|_2^2 \quad \text{subject to} \quad \|x\|_0 \leq \text{sparsity} \end{aligned} \quad (10)$$

There are many different compressed sensing techniques, and Greedy algorithms like Orthogonal Matching Pursuit (OMP) are effective for non-linear recovery [17]. For a Hermitian sensing matrix (A) where $A^{-1} = A^H$, OMP iterates through a defined sparsity (K) and finds the indices (Λ) that maximize the projection of the

$$\begin{array}{c}
\text{Vectorized pulse} \times \text{range-frequencies} \\
\left(\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right) \\
Y \in \mathbb{C}^{ML}
\end{array}
=
\begin{array}{c}
\text{Vectorized pulse} \times \text{range-frequencies} \\
\left(\begin{array}{ccc}
a_{(0,0),0} & \cdots & a_{(0,L-1),K-1} \\
\vdots & & \vdots \\
a_{(M-1,0),0} & \cdots & a_{(M-1,L-1),K-1} \\
\vdots & & \vdots \\
a_{(M-1,L-1),0} & \cdots & a_{(M-1,L-1),K-1}
\end{array} \right) \\
A \in \mathbb{C}^{ML \times K}
\end{array}
\begin{array}{c}
\left(\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right) \\
\text{Vectorized grid points} \\
x \in \mathbb{R}^K
\end{array}$$

Figure 2. Linear system formulation and dimensionality

residual, initialized as the measurement (s), onto the basis vectors formed by A . The maximization of the residual selects the coefficients projected on the basis vectors formed by A , and the step $r \leftarrow s - A_s \hat{\alpha}$ prevents the re-selection of the same indices. The coefficients $\hat{\alpha}$ are the least-squares approximation formed from the selected basis vectors A_s , and they populate the latent image on the support Λ . Algorithm 1 formally describes the OMP algorithm [16]. It is written here in the notation of Section 3 so that the two modifications AR-OMP makes, Algorithm 2, are visible as changes to single lines: the selected atom a_ℓ is replaced by the best of its ambiguous, off-grid candidates, and a cyclic refinement of every selected atom is inserted before the residual update.

Algorithm 1 Orthogonal Matching Pursuit

- | | |
|--|--|
| 1: procedure OMP(A, s, K, τ) | $\triangleright$ dictionary, measurement,
sparsity, noise level |
| 2: $\Lambda \leftarrow \emptyset, A_s \leftarrow [], r \leftarrow s$ | |
| 3: for $i = 1$ to K do | |
| 4: $c \leftarrow A^H r, c_\Lambda \leftarrow 0$ | $\triangleright$ mask atoms already selected |
| 5: $\ell \leftarrow \arg \max_j c_j , \Lambda \leftarrow \Lambda \cup \{\ell\}$ | $\triangleright$ detection |
| 6: $A_s \leftarrow [A_s, a_\ell]$ | $\triangleright$ the selected atom, as it sits on
the grid |
| 7: $\hat{\alpha} \leftarrow (A_s^H A_s)^{-1} A_s^H s$ | |
| 8: $r \leftarrow s - A_s \hat{\alpha}$ | |
| 9: if $\ r\ _2 \leq \tau$ then break | $\triangleright$ residual has reached the noise
level |
| 10: return $\hat{\alpha}, \Lambda$ | |
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3. METHODS

The intention of this section is to walk through the proposed modifications to OMP to handle Doppler ambiguities and off-grid target scatterers. Algorithm 2 illustrates the steps of AR-OMP. AR-OMP is similar in principle to the original OMP algorithm described in Section 2.3 where atoms are selected based on maximization with the residual. However, AR-OMP makes two distinct modifications, assessment of candidate ambiguous atoms from an initial detection in the unambiguous region and off-grid refinement. Section 3.1 reviews the candidate atom selection, SELECTCANDIDATE in Algorithm 2 and detailed in Algorithm 3. Section 3.2 covers the off-grid refinement in the local neighborhoods of the ambiguous scatterers, REFINE and CYCLICREFINE in Algorithm 4.

3.1 Ambiguous candidate atom selection

Because the motion parameters of the target are assumed to be known, the ambiguous doppelganger atoms of an unambiguous target scatterer are known to an extent. The ambiguous scatterers are approximately a Doppler

Algorithm 2 Doppler Ambiguity Refined Orthogonal Matching Pursuit (DAR-OMP)

width away from the position in the unambiguous region. Additionally, the range is approximately known and should be near the range of the unambiguous target scatterer. However, the position of the ambiguous scatterers are not precisely known as a consequence of the non-uniform motion. The non-uniform motion smears the scatterers in the alternate ambiguities. For example, a scatterer located in a -1 ambiguity is sharp with a well-defined mainlobe, but the target is progressively smeared more and more when evaluating the same scatterer at ambiguities 0, +1, +2, etc. The most straightforward method for accounting for the target smearing in other ambiguities is to expand the sensing matrix to include all other ambiguities. However, adding all ambiguities of interest increases the dimensionality of the sensing matrix from K atoms to $N_a * K$ atoms, where N_a is the number of ambiguities. Alternatively a more targeted approach is to evaluate only a select number of candidate atoms in alternate ambiguities. This collection of atoms in each ambiguity will be referred to as a neighborhood. Evaluating only neighborhoods maintains the dimensionality of the sensing matrix during formation and equates to evaluating $K + N_a * K_a$ atoms, where K_a is the number of atoms per ambiguity.

$$\begin{aligned} 2f_D &= \text{prf} \\ 2\omega \times W_x / \lambda &= \text{prf} \quad \text{where for a tabletop} \quad \omega \times W_x = \omega W_x \\ W_x &= (\text{prf})\lambda / 2\omega \end{aligned} \tag{11}$$

$r_k(t_m)$ is the range of the k -th scatterer. For each detected scatterer in the unambiguous region, the range neighborhood is defined as $[y - n_a \Delta_r/2, y + n_a \Delta_r/2]$.

To form the neighborhood, uniformly randomize the positions of K_a candidate atoms within each ambiguity. The randomization distributes the atoms evenly across the neighborhood and prevents poor, systemic initialization of atoms like a line of atoms tracing along a sidelobe. Additionally, to further reduce the number of candidate atoms evaluated, the set of candidate atoms can be refined. The set of candidate atoms forms a dictionary $A_c = [a_{1,1}, \dots, a_{1,K_a}, a_{2,1}, \dots, a_{n_a,k_a}, a_{N_a,K_a}]$ where n_a is the ambiguity index and k_a is the candidate atom index for a given n_a . To discard poorly placed candidate atoms, calculate the correlation of the dictionary A_c with the residual r and retain the top K'_a atoms such that $K'_a < K_a$. Algorithm 3 depicts the candidate atom selection mechanism and refinement.

Algorithm 3 Ambiguous candidate atom selection

1: procedure SELECTCANDIDATE($r, (x_0, y_0), \Delta_c, \Delta_r$)	▷ residual, detected position in the unambiguous region, crossrange search extent, range extent
2: $\mathcal{N} \leftarrow \{[n/2](-1)^n : n = 0, \dots, N_a - 1\}$	▷ ambiguity indices, e.g. $\{-1, 0, 1\}$
3: for $k = 1$ to K_a do	
4: $\delta_k^x \sim \mathcal{U}(-\Delta_c/2, \Delta_c/2), \delta_k^y \sim \mathcal{U}(-\Delta_r/2, \Delta_r/2)$	▷ one offset pair per candidate, shared by every ambiguity
5: $\mathcal{C} \leftarrow \{(x_0 + n_a \bar{W}_x + \delta_k^x, y_0 + \delta_k^y) : n_a \in \mathcal{N}, k = 1, \dots, K_a\}$	
6: $A_c \leftarrow [a(p)]_{p \in \mathcal{C}}$	▷ $N_a K_a$ candidate atoms, never stored in A
7: $\mathcal{C}' \leftarrow$ the K'_a positions of $\mathcal{C}$ with largest $ A_c^H r $	▷ screening, $K'_a = \lfloor N_a K_a / 2 \rfloor$
8: $\gamma^* \leftarrow -\infty$	
9: for $p \in \mathcal{C}'$ do	
10: $\hat{p} \leftarrow \text{REFINE}(r, p)$	▷ Algorithm 4
11: $\gamma \leftarrow a(p)^H r $	
12: if $\gamma > \gamma^*$ then	
13: $\gamma^* \leftarrow \gamma, a^* \leftarrow a(p), p^* \leftarrow \hat{p}$	
14: return a^*, p^*	

3.2 Off-grid neighborhood refinement

After determining the set of candidate atoms $\mathcal{C}'$, apply Newton's method to each atom to refine the off-grid position using the error surface defined in [13]. As derived in Section 2.3, the constrained objective function minimizes $\|y - Ax\|_2^2$ where $S(x, p)$ [13]:

$$\begin{aligned} \|y - Ax\|_2^2 &= y^H y - y^H A x - (A x)^H y + (A x)^H (A x) \\ &= \|y\|_2^2 - 2\Re\{y^H A x\} + \|x\|_2^2 \|A\|_2^2 \end{aligned} \quad (12)$$

Equation

To find the error surface

4. RESULTS

Section 4.1 reviews the effects of non-uniform phase progression on the coherence of the unambiguous and ambiguous atoms. Section 4.2 compares the positional and coefficient root mean squared error relative to other OMP-based algorithms used for off-grid imaging and Doppler ambiguity correction.

Algorithm 4 Off-grid refinement

1: procedure REFINE(r, p)	▷ residual, starting position; R_s Newton steps, B halvings
2: $a \leftarrow a(p), \quad G \leftarrow a^H r ^2 / (a^H a), \quad \hat{p} \leftarrow p$	
3: for $t = 1$ to R_s do	
4: $n \leftarrow a^H a, \quad \alpha \leftarrow a^H r / n, \quad e \leftarrow r - a\alpha$	▷ least-squares amplitude and its residual, $a^H e = 0$
5: $F_i \leftarrow \partial G / \partial p_i$	▷ $p_1 = x, p_2 = y$
6: $H_{ij} \leftarrow \partial^2 G / \partial p_i \partial p_j$	
7: $d \leftarrow H^{-1} F, \quad \text{accepted} \leftarrow \text{false}$	
8: for $b = 0$ to B do	
9: $p' \leftarrow \hat{p} - 2^{-b} d, \quad a' \leftarrow a(p'), \quad G' \leftarrow a'^H r ^2 / (a'^H a')$	
10: if $G' > G$ then	
11: $\hat{p} \leftarrow p', \quad a \leftarrow a', \quad G \leftarrow G', \quad \text{accepted} \leftarrow \text{true}; \quad \text{break}$	▷ backtracking line search
12: if not accepted then break	
13: return $\hat{p}$	
14: procedure CYCLICREFINE($s, \hat{P}, R_c$)	▷ re-refine every atom selected so far
15: for $c = 1$ to R_c do	
16: for $j = 1$ to $ \hat{P} $ do	
17: $r_j \leftarrow s - A_{s, \setminus j} \hat{\alpha}_{\setminus j}$	▷ measurement minus the other atoms
18: $\hat{p}_j \leftarrow \text{REFINE}(r_j, \hat{p}_j), \quad A_{s,j} \leftarrow \vartheta(\hat{p}_j)$	
19: $\hat{\alpha} \leftarrow (A_s^H A_s)^{-1} A_s^H s$	
20: return $\hat{P}, A_s, \hat{\alpha}$	

4.1 Non-uniform sampling coherence

As mentioned in Section 3, non-uniform motion and subsequently, non-uniform phase progression is often seen as a hinderance. Non-uniform motion can introduce phase errors if not approximated and compensated for correctly. However, there is a plethora of literature on translational motion compensation, rotational motion compensation, and autofocus literature capable of target motion estimation. If the motion parameters are assumed to be known, they provide valuable information which can be leveraged to differentiate unambiguous and ambiguous atoms. One measure of the ability to differentiate dictionary atoms is coherence. Coherence is inner product between two atoms, $a_i^H a_j$, and represents the similarity between those vectors. Mutual coherence is the maximum coherence between any pair of atoms and has been used as a measure of atom differentiation in CS [cite]. The mutual coherence is the coherence between atoms within the sensing matrix, and is defined as:

$$M = \max_{0 \leq i \neq j \leq K-1} |a_i^H a_j| \quad (13)$$

To characterize and quantify the amount of help non-uniform phase progression provides, the mutual coherence of the sensing matrix was calculated as a function of the motion parameters.

4.2 RMSE

To test the effectiveness of Doppler ambiguity and ,

4.3 Noise assessment

5. DISCUSSION

5.1 Degradation for closely-space scatterers

6. CONCLUSION

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